

Data Understanding & Data Quality Assessment (Excel)

Objective:

The objective of this phase is to understand the retail sales dataset and assess its quality before performing data cleaning in Power

Query. The dataset was examined in Microsoft Excel to identify missing values, duplicate records, incorrect data types, invalid values, and other data quality issues that could affect analysis.

Tasks Performed:

Opened the Orders and Returns datasets in Microsoft Excel.

Verified the number of rows and columns in each sheet.

Identified missing (NULL) values.

Checked for duplicate records.

Validated Order Date and Ship Date formats.

Verified data types for all columns.

Identified invalid values in Quantity and Discount columns.

Documented all data quality issues before cleaning.

Data Quality Report:

Issue	Column	Count	Action Required
Missing Values	Customer Name	31	Removed affected rows during cleaning
Missing Values	State	31	Removed affected rows (same records as Customer Name)
Missing Values	Category	31	Removed affected rows (same records as Customer Name)
Missing Values	Discount	4	Removed affected rows
Duplicate Records	Returns	3	Removed duplicate rows
Invalid Values	Quantity	6	Corrected negative values (-3 to 3)
Invalid Values	Discount	6	Replaced invalid values (abc, -5, 120, 55, 65, etc.) and converted the column to Decimal Number
Data Types	Order Date, Ship Date	All records	Converted to Date data type

Summary:

The dataset contained a small number of missing values, duplicate records, and invalid values. These issues were identified during the Excel data quality assessment and were addressed later in Power Query. After cleaning, the dataset was ready for transformation, modeling, and dashboard development.